

WEEK – 5 DBMS TASK

PAYMENT TRANSACTION MANAGEMENT SYSTEM

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TASK DESCRIPTION

1. Create Account and Transaction tables.
2. Store account holder and transaction details.
3. Track credit and debit transactions.
4. Analyze transaction types and balances.
5. Generate account-wise transaction reports.

ABSTRACT

The Payment Transaction Management System is designed to manage and monitor financial transactions efficiently. It stores customer account information, transaction records, payment methods, and transaction statuses. The system helps organizations maintain accurate financial data, track transaction history, and generate detailed reports. It improves transparency, security, and efficiency in payment processing.

INTRODUCTION

In modern businesses, handling payment transactions accurately is essential. A Payment Transaction Management System provides a structured approach for recording, processing, and monitoring payments. The system stores customer information, transaction details, payment methods, and transaction status. It enables quick access to financial records and supports effective decision-making through analytical reports.

OBJECTIVES

- Maintain customer payment records.
- Store transaction details securely.
- Monitor successful and failed transactions.
- Generate financial reports.
- Improve transaction management efficiency.

RESULT

The system successfully records customer transactions, tracks payment status, and generates transaction reports. It provides a reliable platform for managing payment-related activities and analyzing transaction performance.

CONCLUSION

The Payment Transaction Management System offers an efficient solution for managing financial transactions. It ensures data accuracy, simplifies transaction tracking, and supports better financial management through detailed reporting and analysis.

```
-- =====  
-- PAYMENT TRANSACTION MANAGEMENT SYSTEM  
-- NEW DATABASE DESIGN  
-- =====
```

```
CREATE DATABASE PaymentTransactionDB;  
USE PaymentTransactionDB;
```

```
-- =====  
-- 1. CREATE ACCOUNT TABLE  
-- =====
```

```
CREATE TABLE Account (  
  
    account_id INT PRIMARY KEY,  
    customer_name VARCHAR(100) NOT NULL,  
    account_type VARCHAR(30),  
    mobile_no VARCHAR(15),  
    city VARCHAR(50)  
);
```

```
-- =====  
-- 2. CREATE TRANSACTION TABLE  
-- =====
```

```
CREATE TABLE TransactionDetails (  
    transaction_id INT PRIMARY KEY,  
    account_id INT,  
    transaction_type VARCHAR(20),  
    payment_method VARCHAR(30),  
    transaction_date DATE,  
    amount DECIMAL(10,2),  
    status VARCHAR(20),  
    FOREIGN KEY (account_id)  
    REFERENCES Account(account_id)  
);
```

```
-- =====
-- 3. INSERT ACCOUNT DETAILS
-- =====
```

```
INSERT INTO Account VALUES
(101,'Rahul Sharma','Savings','9876543210','Chennai'),
(102,'Priya Nair','Current','9876543211','Bangalore'),
(103,'Arjun Kumar','Savings','9876543212','Coimbatore'),
(104,'Sneha Devi','Current','9876543213','Madurai'),
(105,'Vignesh R','Savings','9876543214','Salem');
```

```
-- =====
-- 4. INSERT TRANSACTION DETAILS
-- =====
```

```
INSERT INTO TransactionDetails VALUES
(1001,101,'Credit','UPI','2026-09-01',5000,'Completed'),
(1002,102,'Debit','Credit Card','2026-09-02',2500,'Completed'),
(1003,103,'Credit','Net Banking','2026-09-03',8000,'Pending'),
(1004,104,'Debit','UPI','2026-09-04',1200,'Failed'),
(1005,105,'Credit','Debit Card','2026-09-05',6500,'Completed'),
(1006,101,'Debit','Cash','2026-09-06',3000,'Completed');
```

```
-- =====
-- 5. DISPLAY ALL ACCOUNTS
-- =====
```

```
SELECT * FROM Account;
```

```
-- =====
-- 6. DISPLAY ALL TRANSACTIONS
-- =====
```

```
SELECT * FROM TransactionDetails;
```

```
-- =====
-- 7. COMPLETED TRANSACTIONS
-- =====
```

```
SELECT *
FROM TransactionDetails
WHERE status='Completed';
```

```
-- =====
-- 8. PENDING TRANSACTIONS
-- =====
```

```
SELECT *
FROM TransactionDetails
WHERE status='Pending';
```

```
-- =====
-- 9. FAILED TRANSACTIONS
-- =====
```

```
SELECT *
FROM TransactionDetails
WHERE status='Failed';
```

```
-- =====
-- 10. ACCOUNT WITH TRANSACTION DETAILS
-- =====
```

```
SELECT
a.customer_name,
t.transaction_id,
t.transaction_type,
t.payment_method,
t.amount,
t.status
FROM Account a
JOIN TransactionDetails t
ON a.account_id=t.account_id;
```

```
-- =====
-- 11. TOTAL TRANSACTION AMOUNT
-- =====
```

```
SELECT SUM(amount) AS Total_Amount
FROM TransactionDetails;
```

```
-- =====
-- 12. TRANSACTION COUNT BY PAYMENT METHOD
-- =====
```

```
SELECT payment_method,
COUNT(*) AS Total_Transactions
FROM TransactionDetails
GROUP BY payment_method;
```

```
-- =====
-- 13. HIGHEST TRANSACTION
-- =====
```

```
SELECT *
```

```
FROM TransactionDetails
ORDER BY amount DESC
LIMIT 1;
```

```
-- =====
-- 14. CUSTOMER-WISE TRANSACTION REPORT
-- =====
```

```
SELECT
a.customer_name,
COUNT(t.transaction_id) AS Total_Transactions,
SUM(t.amount) AS Total_Amount
FROM Account a
JOIN TransactionDetails t
ON a.account_id=t.account_id
GROUP BY a.customer_name;
```

```
-- =====
-- 15. DAILY TRANSACTION REPORT
-- =====
```

```
SELECT
transaction_date,
COUNT(*) AS Transaction_Count,
SUM(amount) AS Daily_Total
FROM TransactionDetails
GROUP BY transaction_date
ORDER BY transaction_date;
```

